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RBF-ESO Backstepping Control for Multi-UAV Cooperative Payload Transportation

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Abstract

Cooperative payload transportation with multiple unmanned aerial vehicles (UAVs) is challenging because the payload dynamics are strongly coupled with cable constraints, nonlinear rotational motion, allocation errors, and external disturbances. This paper proposes an RBF-ESO backstepping control framework for robust payload-level trajectory tracking. A nominal backstepping controller is first constructed to generate the desired payload force and moment commands. A radial basis function (RBF) neural network is then introduced to approximate unmodeled nonlinear dynamics online, while an extended state observer (ESO) estimates the remaining lumped disturbances for feedforward compensation. To evaluate the contribution of each component, the proposed controller is tested through backstepping-family ablation studies under nominal and disturbed conditions. Simulation results show that the combined RBF-ESO compensation improves both tracking accuracy and disturbance rejection. The ablation results demonstrate that integrating neural approximation and observer-based disturbance estimation within a backstepping structure provides a balanced and robust control strategy for multi-UAV cooperative payload transportation.

Keywords

multi-UAV cooperative transportation; payload transportation; backstepping control; RBF neural network; extended state observer

Introduction

UAV technology has developed rapidly in recent years and is now widely used in aerial imaging, agriculture, industry, and material delivery. Recently, increasing attention has been directed toward multi-UAV systems. Compared with single-UAV platforms, multi-UAV cooperative systems provide higher payload capacity, greater flexibility, and improved fault tolerance. Among these applications, multi-UAV cooperative payload transportation has attracted significant interest due to its potential for transporting heavy and large-sized materials Fink et al. (2011); Lee et al. (2013c); Estevez et al. (2024).

However, cooperative payload transportation remains a challenging problem in aerial robotics. The presence of flexible cables, coupled motion constraints, and strong interactions among multiple UAVs and the payload leads to a high-dimensional, nonlinear, and tightly coupled dynamical system. In practice, these couplings introduce significant difficulties in both controller design and stability analysis, especially when precise trajectory tracking and attitude regulation are required simultaneously Goodarzi et al. (2015); Sreenath et al. (2013); Lee (2018). Although a complete geometric dynamic model has been established in prior work Lee et al. (2013a); Lee (2014), achieving robust and high-accuracy control under external disturbances, parameter uncertainties, and unmodeled dynamics is still nontrivial. Therefore, developing control strategies that combine accurate modeling with strong disturbance-rejection capability remains an important research direction for cooperative multi-UAV payload transportation.

To address these challenges, a variety of control strategies have been developed in the literature. Nonlinear methods, such as backstepping and sliding-mode control, have been widely adopted for UAV systems because of their ability to handle strong nonlinearities Bouabdallah and Siegwart (2005); Madani and Benallegue (2008); Thanh et al. (2022). Nevertheless, these approaches often depend on relatively accurate system models, and their performance can degrade when significant parameter

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uncertainties and unmodeled dynamics are present. To mitigate these limitations, adaptive control schemes and neural-network-based methods have been introduced to approximate unknown dynamics online Sanner and Slotine (1992); Park and Sandberg (1991); Shen and Xu (2021). In addition, ESO and ADRC methods provide a practical observer-based route for estimating lumped disturbances and improving robustness Han (2009); Gao (2003); Madoński and Herman (2015); Wang et al. (2022). These developments motivate the use of learning-based compensation terms together with robust feedback to improve tracking performance under modeling uncertainty.

In this paper, we develop an RBF-ESO backstepping controller for multi-UAV cooperative payload transportation. Backstepping is adopted as the nominal nonlinear tracking controller because it provides a recursive error-stabilization structure for the payload translation and attitude channels. An RBF neural network is then introduced to approximate unmodeled nonlinear dynamics online Broomhead and Lowe (1988); Park and Sandberg (1991); Shen and Xu (2021), while an ESO is used to estimate the residual lumped disturbance that remains after neural approximation Han (2009); Madoński and Herman (2015); Gu et al. (2022). The resulting compensation terms are incorporated directly into the payload force and moment commands.

The main contribution of this paper is a simulation-oriented ablation framework for evaluating the proposed composite controller. The framework compares plain backstepping, RBF-augmented backstepping, ESO-augmented backstepping, and the combined RBF-ESO backstepping controller under both no-disturbance and disturbance-injection cases. This organization makes it possible to identify the individual roles of RBF approximation and ESO disturbance estimation within the same cooperative transportation task.

System Modeling

The dynamic model used in this paper is based on the geometric formulation proposed by Lee et al. for multiple quadrotor UAVs transporting a cable-suspended rigid body Lee et al. (2013a,c). In that framework, a cooperative aerial transportation system consists of N quadrotors carrying a rigid payload through massless, taut, inextensible

cables. The payload dynamics and individual quadrotor dynamics are derived separately and are coupled through kinematic constraints.

Coordinate Frames and State Representation

We define three primary coordinate frames:

Inertial Frame ($\mathcal{I}$) A fixed Earth frame defined by basis vectors $\{e_1, e_2, e_3\}$, where $e_3 = [0, 0, 1]^T$ denotes the positive vertical axis; gravity is represented as $+ge_3$ under the NED convention.

Payload Body Frame ($\mathcal{L}$) Attached to the payload center of mass (CoM), with orientation $\mathbf{R}_L \in SO(3)$ relative to $\mathcal{I}$.

Quadrotor Body Frame ($\mathcal{B}_i$) Attached to the CoM of the i-th quadrotor ($i = 1 \dots N$), with orientation $\mathbf{R}_i \in SO(3)$ relative to $\mathcal{I}$.

For both the payload and each quadrotor, the dynamic variables are position $\mathbf{p}$, linear velocity $\mathbf{v}$, attitude matrix $\mathbf{R}$, and angular velocity $\mathbf{\Omega}$. Here, $\mathbf{p}$ and $\mathbf{v}$ are expressed in the inertial frame, $\mathbf{R} \in SO(3)$ denotes attitude, and $\mathbf{\Omega}$ is expressed in the corresponding body frame. Subscripts denote object indices: L for the payload, i for the i-th quadrotor (or cable), and ext for external disturbances.

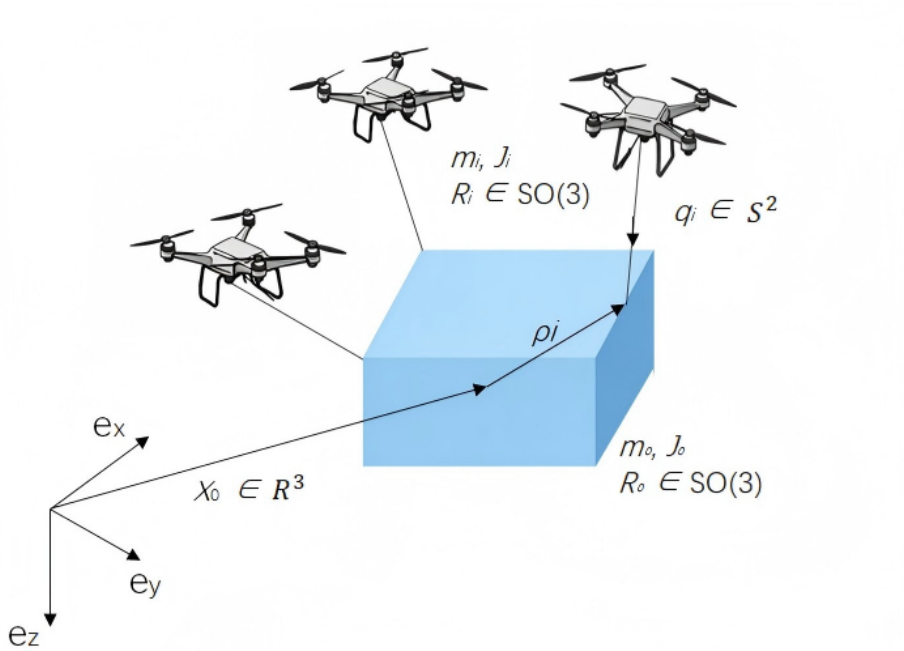


Figure 1. The UAVs and payload geometric structure.

Kinematic Constraints

The payload is connected to the i-th quadrotor by a taut, inextensible cable of length l_i. Let $\boldsymbol{\rho}_i$ be the vector of the connection point on the load in the $\mathcal{L}$ frame, then the geometric relationship follows the equation below.

$$\mathbf{p}_i = \mathbf{p}_L + \mathbf{R}_L \boldsymbol{\rho}_i - l_i \mathbf{q}_i \tag{1}$$

where $\mathbf{q}_i \in \mathbb{S}^2$ is the unit vector representing the cable direction, pointing from the quadrotor to the payload.

Then differentiating with respect to time yields the velocity relationship:

$$\mathbf{v}_i = \mathbf{v}_L + \mathbf{R}_L(\boldsymbol{\Omega}_L \times \boldsymbol{\rho}_i) - l_i \boldsymbol{\omega}_i \times \mathbf{q}_i \tag{2}$$

Here, the cable angular velocity $\boldsymbol{\omega}_{cable,i}$ satisfies $\dot{\mathbf{q}}_i = \boldsymbol{\omega}_i \times \mathbf{q}_i$.

Payload Dynamics

We consider a 6-DOF rigid body with mass m_L and inertia tensor $\mathbf{J}_L$,and derive the equations of motion using the Newton-Euler method..

Translational Dynamics The total force acting on the payload is composed of gravity, the tension forces $\mathbf{T}_i$ from all N cables, and an external disturbance force $\mathbf{f}_{ext}$:

$$m_L \dot{\mathbf{v}}_L = m_L g \mathbf{e}_3 + \sum_{i=1}^{N} \mathbf{T}_i + \mathbf{f}_{ext} \tag{3}$$

where $\mathbf{T}_i = -\mu_i \mathbf{q}_i$ is the tension vector, with $\mu_i > 0$ being the tension magnitude, and $\mathbf{f}_{ext} \in \mathbb{R}^3$ denotes the lumped external disturbance force.

Rotational Dynamics The rotational motion is driven by the moments generated by the cable tensions applied at lever arms $\boldsymbol{\rho}_i$:

$$\mathbf{J}_L \dot{\boldsymbol{\Omega}}_L = -\boldsymbol{\Omega}_L \times (\mathbf{J}_L \boldsymbol{\Omega}_L) + \sum_{i=1}^{N} \boldsymbol{\rho}_i \times (\mathbf{R}_L^T \mathbf{T}_i) + \boldsymbol{\tau}_{ext} \tag{4}$$

where $\boldsymbol{\tau}_{ext}$ represents external disturbance torques.

Quadrotor Dynamics

Each quadrotor is treated as an independent rigid body subject to its own thrust, gravity, and the reaction force from the cable tension ($-\mathbf{T}_i$).

Translational Dynamics

$$m_i \dot{\mathbf{v}}_i = m_i g \mathbf{e}_3 - f_i \mathbf{R}_i \mathbf{e}_3 - \mathbf{T}_i \tag{5}$$

where f_i is the total scalar thrust generated by the i-th quadrotor rotors, and $-\mathbf{R}_i \mathbf{e}_3$ represents the thrust direction in the inertial frame.

Rotational Dynamics

$$\mathbf{J}_i \dot{\boldsymbol{\Omega}}_i = -\boldsymbol{\Omega}_i \times (\mathbf{J}_i \boldsymbol{\Omega}_i) + \boldsymbol{\tau}_i \tag{6}$$

where $\boldsymbol{\tau}_i = [\tau_\phi, \tau_\theta, \tau_\psi]^T$ are the control torques in the body frame.

Hierarchical Control Design

To address uncertainties in payload dynamics and external disturbances, we propose an adaptive robust control scheme that integrates backstepping control, radial basis function (RBF) neural-network approximation, and extended state observer (ESO) disturbance estimation while following the

geometric payload-transport control framework in Lee et al. (2013a); Lee (2018). The backstepping component generates the nominal payload force and moment commands, the RBF network compensates for unmodeled nonlinear dynamics Sanner and Slotine (1992); Park and Sandberg (1991), and the ESO estimates residual disturbances that are not fully captured by the neural approximation Han (2009); Madoński and Herman (2015).

The overall signal flow of the proposed hierarchical controller is shown in Fig. 2. The payload tracking errors are first processed by the nominal backstepping structure, while the RBF and ESO modules provide online compensation for model uncertainty and residual disturbances. The resulting payload-level force and moment commands are then distributed through the cable-force allocation layer and converted into executable quadrotor thrust and attitude commands.

State and Error Definitions

The payload translational state is described by the position $\mathbf{p}_L \in \mathbb{R}^3$ and linear velocity $\mathbf{v}_L \in \mathbb{R}^3$, with desired trajectories $\mathbf{p}_{L,d}$ and $\mathbf{v}_{L,d}$. The payload attitude can be equivalently represented by the rotation matrix $\mathbf{R}_L \in SO(3)$ for geometric modeling. However, the implemented RBF-adaptive Backstepping-ESO controller uses the Euler-angle coordinates of the payload attitude. Let

$$\boldsymbol{\Theta}_L = \begin{bmatrix} \phi_L & \theta_L & \psi_L \end{bmatrix}^\top, \qquad \boldsymbol{\Theta}_{L,d} = \begin{bmatrix} \phi_{L,d} & \theta_{L,d} & \psi_{L,d} \end{bmatrix}^\top, \tag{7}$$

denote the current and desired payload Euler-angle vectors, respectively. The payload tracking errors used in the controller are therefore defined as

$$\begin{aligned} \mathbf{e}_{p_L} &= \mathbf{p}_L - \mathbf{p}_{L,d}, \\ \mathbf{e}_{v_L} &= \mathbf{v}_L - \mathbf{v}_{L,d}, \\ \mathbf{e}_{\Theta_L} &= \boldsymbol{\Theta}_L - \boldsymbol{\Theta}_{L,d}. \end{aligned} \tag{8}$$

Based on these definitions, the composite input vector of the RBF network is chosen as

$$\boldsymbol{\xi} = \begin{bmatrix} \mathbf{e}_{p_L} \\ \mathbf{e}_{\Theta_L} \end{bmatrix} \in \mathbb{R}^6. \tag{9}$$

RBF Neural Network Approximation

RBF networks are widely used for nonlinear function approximation because Gaussian basis functions yield a linearly parameterized

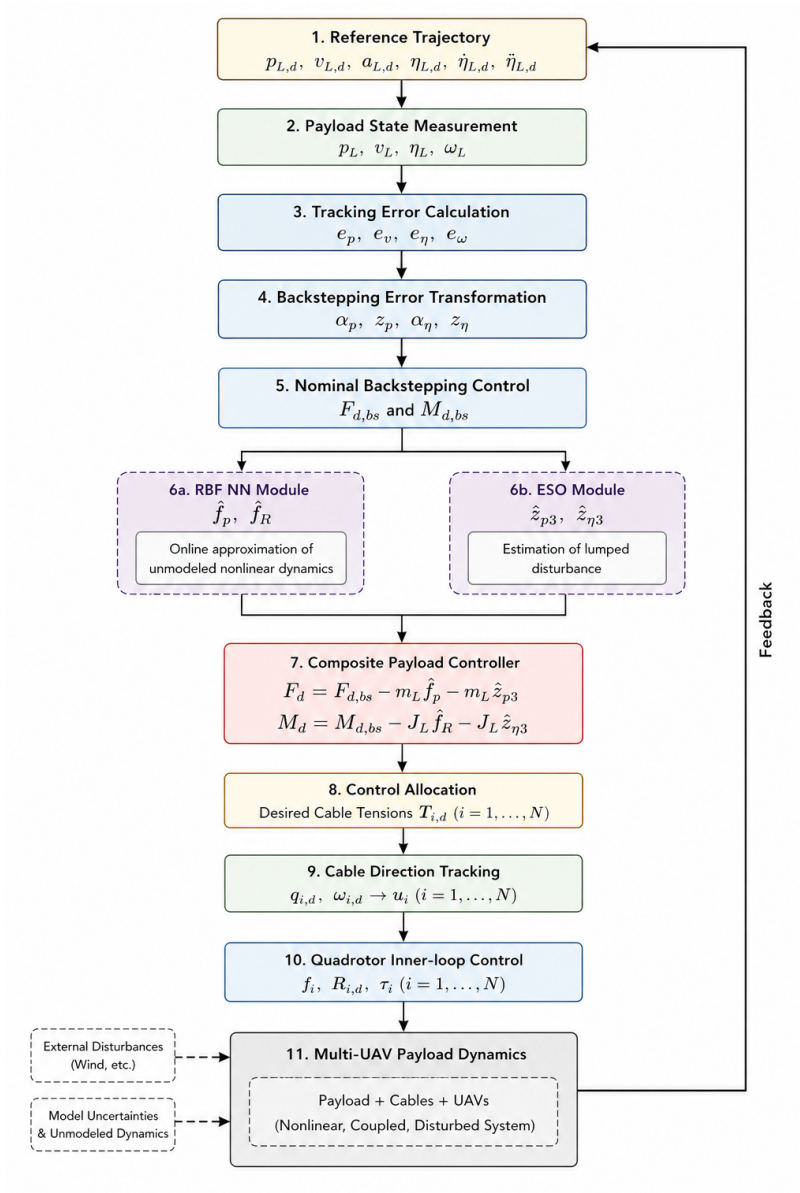


Figure 2. Pipeline of the proposed RBF-ESO backstepping control framework for multi-UAV cooperative payload transportation.

approximator with universal approximation capability Broomhead and Lowe (1988); Park and Sandberg (1991). In this project, the RBF approximation is not implemented as a single global network. Instead, six scalar Gaussian RBF networks are used in parallel: three for payload translation and three for payload attitude. All six channels share the same composite input vector $\boldsymbol{\xi} \in \mathbb{R}^6$ defined in (9), while each channel has its own adaptive weight vector. This structure matches the controller implementation, where five basis nodes are assigned to each controlled degree of freedom, resulting in a total of $6 \times 5 = 30$ adaptive parameters.

The channel-wise RBF structure used for online uncertainty approximation is illustrated in Fig. 3. The shared tracking-error input is mapped through Gaussian basis functions, and the resulting channel outputs are injected into the translational and rotational payload controllers as adaptive compensation terms.

For the j-th basis node, the Gaussian activation is defined as

$$h_j(\boldsymbol{\xi}) = \exp\left(-\frac{\|\boldsymbol{\xi} - \mathbf{c}_j\|^2}{2b_j^2}\right), \qquad j = 1, \ldots, 5, \tag{10}$$

where $\mathbf{c}_j \in \mathbb{R}^6$ is the center of the j-th node and $b_j > 0$ is its width. To improve numerical robustness, the implementation normalizes the basis functions according to

$$\bar{\mathbf{h}}(\boldsymbol{\xi}) = \frac{\mathbf{h}(\boldsymbol{\xi})}{\sum_{j=1}^{5} h_j(\boldsymbol{\xi})}, \qquad \mathbf{h}(\boldsymbol{\xi}) = \begin{bmatrix} h_1(\boldsymbol{\xi}) & \cdots & h_5(\boldsymbol{\xi}) \end{bmatrix}^\top, \tag{11}$$

whenever $\sum_{j=1}^{5} h_j(\boldsymbol{\xi})$ is non-negligible.

For the translational channels, let $\hat{\boldsymbol{\theta}}_{p,i} \in \mathbb{R}^5$, $i = 1, 2, 3$, denote the adaptive weight vectors for the x-, y-, and z-directions. The RBF outputs are

$$\hat{f}_{p,i}(\boldsymbol{\xi}) = \hat{\boldsymbol{\theta}}_{p,i}^\top \bar{\mathbf{h}}(\boldsymbol{\xi}), \qquad i = 1, 2, 3, \tag{12}$$

and they are stacked as

$$\hat{\mathbf{f}}_p(\boldsymbol{\xi}) = \begin{bmatrix} \hat{f}_{p,1}(\boldsymbol{\xi}) & \hat{f}_{p,2}(\boldsymbol{\xi}) & \hat{f}_{p,3}(\boldsymbol{\xi}) \end{bmatrix}^\top. \tag{13}$$

In the implemented force controller, $\hat{\mathbf{f}}_p(\boldsymbol{\xi})$ is interpreted as an estimate of acceleration-level uncertainty in the payload translational dynamics, which is why it appears in the control law multiplied by the payload mass m_L.

Similarly, for the rotational channels, let $\hat{\boldsymbol{\theta}}_{R,i} \in \mathbb{R}^5$, $i = 1, 2, 3$, denote the adaptive weights for roll, pitch, and yaw. The corresponding outputs are

$$\hat{f}_{R,i}(\boldsymbol{\xi}) = \hat{\boldsymbol{\theta}}_{R,i}^{\top} \bar{\mathbf{h}}(\boldsymbol{\xi}), \qquad i = 1, 2, 3, \tag{14}$$

and the rotational compensation vector is written as

$$\hat{\mathbf{f}}_R(\boldsymbol{\xi}) = \begin{bmatrix} \hat{f}_{R,1}(\boldsymbol{\xi}) & \hat{f}_{R,2}(\boldsymbol{\xi}) & \hat{f}_{R,3}(\boldsymbol{\xi}) \end{bmatrix}^{\top}. \tag{15}$$

These two vectors are then injected into the translational and rotational backstepping controllers as adaptive compensation terms.

The adaptive weights are updated online using channel-wise Lyapunov-based laws. In the implementation, the update law uses the same six-dimensional RBF input vector $\boldsymbol{\xi}$ defined in (9).

The continuous-time adaptive law for each channel is implemented in the Lyapunov-based RBF adaptation form commonly used in adaptive neural control Sanner and Slotine (1992); Patino and Liu (2000):

$$\dot{\hat{\boldsymbol{\theta}}}_i = \gamma_{i,\text{eff}}\, \bar{\mathbf{h}}(\boldsymbol{\xi})\, \boldsymbol{\xi}^{\top} \mathbf{P}_i \mathbf{B}_i - \kappa\, \gamma_{i,\text{eff}}\, \hat{\boldsymbol{\theta}}_i, \tag{16}$$

where $\bar{\mathbf{h}}(\boldsymbol{\xi})$ is the normalized RBF basis driven by the six-dimensional payload tracking-error input. The matrices $\mathbf{P}_i$ and $\mathbf{B}_i$ select and weight the corresponding channel error in the adaptive update, while the input vector itself remains $\boldsymbol{\xi}$. In addition, $\kappa > 0$ is the leakage gain introduced to suppress parameter drift in the presence of approximation error, and $\gamma_{i,\text{eff}}$ is an error-dependent effective adaptation gain Patino and Liu (2000). In the actual code, this gain is reduced inside a dead-zone region according to

$$\gamma_{i,\text{eff}} = \gamma_i \begin{cases} \dfrac{\|\boldsymbol{\xi}\|}{\sigma_i}, & \|\boldsymbol{\xi}\| < \sigma_i, \\ 1, & \|\boldsymbol{\xi}\| \geq \sigma_i, \end{cases} \tag{17}$$

where σ_i is the dead-zone threshold. Equivalently, $\gamma_{i,\text{eff}} = \gamma_i \min(1, \|\boldsymbol{\xi}\|/\sigma_i)$. In the current implementation, $\gamma_i = 80$ and $\sigma_i = 0.1$ for the x-, y-, and attitude channels, whereas the z-translation channel uses $\gamma_i = 20$ and $\sigma_i = 0.05$. The discrete implementation uses forward Euler integration,

$$\hat{\boldsymbol{\theta}}_i(k+1) = \hat{\boldsymbol{\theta}}_i(k) + \Delta t\, \dot{\hat{\boldsymbol{\theta}}}_i(k), \tag{18}$$

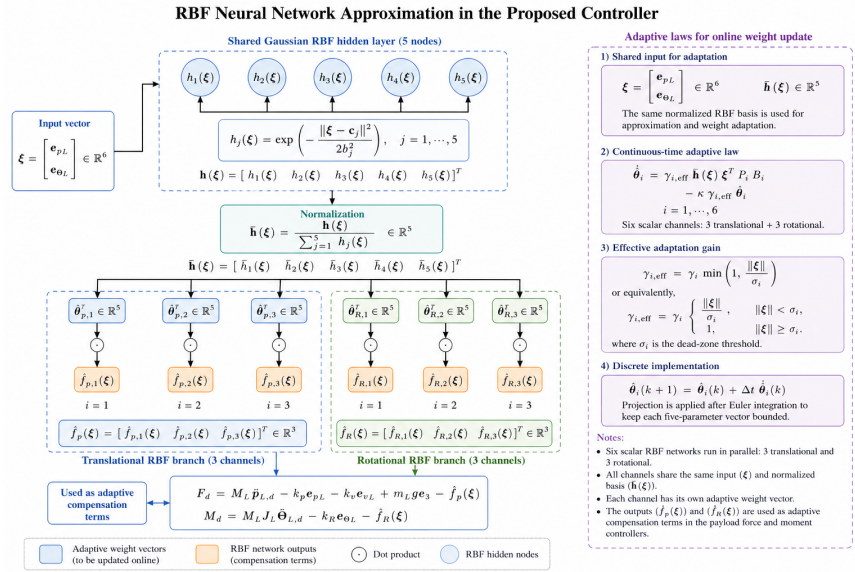


Figure 3. Channel-wise RBF neural network used for payload uncertainty approximation.

followed by parameter projection to enforce elementwise bounds and a norm bound on each five-parameter vector. In the current implementation, the z-translation channel is updated more conservatively than the x-, y-, and attitude channels because it uses a smaller nominal adaptation gain and a tighter dead-zone threshold.

ESO-Based Disturbance Estimation

Although RBF neural network can effectively estimate the unknown disturbance, the existence of model residual uncertainties and external disturbances may still degrade the control accuracy. To further improve the robustness of the system, an extended state observer (ESO) is introduced to estimate the residual disturbance online, following the disturbance-estimation idea of ADRC/ESO-based control Han (2009); Gao (2003); Madoński and Herman (2015); Gu et al. (2022).

For the translational channel, the payload dynamics can be rewritten in acceleration form as

$$\ddot{\mathbf{p}}_L = \mathbf{F}_p + \mathbf{d}_p, \tag{19}$$

where $\mathbf{p}_L$ is the payload position, $\mathbf{F}_p \in \mathbb{R}^3$ denotes the known control-related acceleration term, and $\mathbf{d}_p \in \mathbb{R}^3$ represents the translational lumped disturbance including modeling errors and external perturbations.

Define the augmented states as

$$\mathbf{z}_{p1} = \mathbf{p}_L, \qquad \mathbf{z}_{p2} = \dot{\mathbf{p}}_L, \qquad \mathbf{z}_{p3} = \mathbf{d}_p. \tag{20}$$

Then, the corresponding ESO is designed as

$$\begin{cases} \dot{\hat{\mathbf{z}}}_{p1} = \hat{\mathbf{z}}_{p2} + \beta_{p1}\left(\mathbf{z}_{p1} - \hat{\mathbf{z}}_{p1}\right), \\ \dot{\hat{\mathbf{z}}}_{p2} = \hat{\mathbf{z}}_{p3} + \mathbf{F}_p + \beta_{p2}\left(\mathbf{z}_{p1} - \hat{\mathbf{z}}_{p1}\right), \\ \dot{\hat{\mathbf{z}}}_{p3} = \beta_{p3}\left(\mathbf{z}_{p1} - \hat{\mathbf{z}}_{p1}\right), \end{cases} \tag{21}$$

where $\hat{\mathbf{z}}_{p1}$, $\hat{\mathbf{z}}_{p2}$, and $\hat{\mathbf{z}}_{p3}$ are the estimates of the payload position, velocity, and translational lumped disturbance, respectively.

Similarly, for the attitude channel, the dynamics can be expressed as

$$\ddot{\boldsymbol{\eta}}_L = \mathbf{F}_\eta + \mathbf{d}_\eta, \tag{22}$$

where $\boldsymbol{\eta}_L \in \mathbb{R}^3$ denotes the payload attitude vector, $\mathbf{F}_\eta \in \mathbb{R}^3$ is the known control-related angular-acceleration term, and $\mathbf{d}_\eta \in \mathbb{R}^3$ is the lumped disturbance in the attitude channel.

Define the augmented states as

$$\mathbf{z}_{\eta 1} = \boldsymbol{\eta}_L, \qquad \mathbf{z}_{\eta 2} = \dot{\boldsymbol{\eta}}_L, \qquad \mathbf{z}_{\eta 3} = \mathbf{d}_\eta. \tag{23}$$

The ESO for the attitude channel is then given by

$$\begin{cases} \dot{\hat{\mathbf{z}}}_{\eta 1} = \hat{\mathbf{z}}_{\eta 2} + \beta_{\eta 1}\left(\mathbf{z}_{\eta 1} - \hat{\mathbf{z}}_{\eta 1}\right), \\ \dot{\hat{\mathbf{z}}}_{\eta 2} = \hat{\mathbf{z}}_{\eta 3} + \mathbf{F}_\eta + \beta_{\eta 2}\left(\mathbf{z}_{\eta 1} - \hat{\mathbf{z}}_{\eta 1}\right), \\ \dot{\hat{\mathbf{z}}}_{\eta 3} = \beta_{\eta 3}\left(\mathbf{z}_{\eta 1} - \hat{\mathbf{z}}_{\eta 1}\right). \end{cases} \tag{24}$$

Based on the estimated disturbances, the compensation terms can be constructed as

$$\mathbf{u}_{ESO,p} = -\hat{\mathbf{z}}_{p3}, \qquad \mathbf{u}_{ESO,\eta} = -\hat{\mathbf{z}}_{\eta 3}. \tag{25}$$

By incorporating these terms into the backstepping control law, the adverse effects of uncertainties and disturbances can be effectively attenuated, thereby improving the overall robustness and tracking performance of the cooperative transportation system.

Payload Control Law

Using the tracking-error definitions, RBF approximation terms, and ESO disturbance estimates introduced above, the payload-layer controller computes the desired force $\mathbf{F}_d \in \mathbb{R}^3$ in the inertial frame and the desired moment $\mathbf{M}_d \in \mathbb{R}^3$ applied to the payload. This subsection therefore focuses on how these previously defined compensation terms are incorporated into the force and moment control laws.

Translational Control (Force Generation) Define the position and velocity tracking errors as

$$\mathbf{e}_{p_L} = \mathbf{p}_L - \mathbf{p}_{L,d}, \qquad \mathbf{e}_{v_L} = \mathbf{v}_L - \dot{\mathbf{p}}_{L,d}. \tag{26}$$

Following the Backstepping design, introduce the virtual control

$$\boldsymbol{\alpha}_p = -\mathbf{K}_{p1}\mathbf{e}_{p_L} + \dot{\mathbf{p}}_{L,d}, \tag{27}$$

where $\mathbf{K}_{p1} \succ \mathbf{0}$ is a diagonal gain matrix. Then define the second-layer error as

$$\mathbf{z}_p = \mathbf{v}_L - \boldsymbol{\alpha}_p = \mathbf{e}_{v_L} + \mathbf{K}_{p1}\mathbf{e}_{p_L}. \tag{28}$$

The nominal Backstepping force is designed as

$$\mathbf{F}_{d,bs} = m_L(\ddot{\mathbf{p}}_{L,d} - \mathbf{K}_{p1}\mathbf{e}_{v_L} - \mathbf{K}_{p2}\mathbf{z}_p + g\mathbf{e}_3), \tag{29}$$

where $\mathbf{K}_{p2} \succ \mathbf{0}$ is a diagonal gain matrix.

The RBF translational estimate $\hat{\mathbf{f}}_p(\boldsymbol{\xi})$ in (9)–(16) and the ESO residual-disturbance estimate $\hat{\mathbf{z}}_{p3}$ in (20)–(21) are then added as compensation terms. The translational control law is

$$\mathbf{F}_d = \mathbf{F}_{d,bs} - m_L\hat{\mathbf{f}}_p(\boldsymbol{\xi}) - m_L\hat{\mathbf{z}}_{p3}. \tag{30}$$

Here, $\hat{\mathbf{z}}_{p3}$ denotes the componentwise translational disturbance estimate provided by the ESO.

Rotational Control (Moment Generation) Define the payload attitude and angular-velocity tracking errors as

$$\mathbf{e}_{\eta_L} = \boldsymbol{\eta}_L - \boldsymbol{\eta}_{L,d}, \qquad \mathbf{e}_{\omega_L} = \boldsymbol{\omega}_L - \dot{\boldsymbol{\eta}}_{L,d}, \tag{31}$$

where $\boldsymbol{\eta}_L$ denotes the payload attitude vector.

Similarly, define the virtual control for the rotational channel as

$$\boldsymbol{\alpha}_{\eta} = -\mathbf{K}_{\eta 1}\mathbf{e}_{\eta_L} + \dot{\boldsymbol{\eta}}_{L,d}, \tag{32}$$

and the second-layer error as

$$\mathbf{z}_{\eta} = \boldsymbol{\omega}_L - \boldsymbol{\alpha}_{\eta} = \mathbf{e}_{\omega_L} + \mathbf{K}_{\eta 1}\mathbf{e}_{\eta_L}, \tag{33}$$

where $\mathbf{K}_{\eta 1} \succ \mathbf{0}$.

The nominal Backstepping moment is designed as

$$\mathbf{M}_{d,bs} = \mathbf{J}_L(\ddot{\boldsymbol{\eta}}_{L,d} - \mathbf{K}_{\eta 1}\mathbf{e}_{\omega_L} - \mathbf{K}_{\eta 2}\mathbf{z}_{\eta}), \tag{34}$$

where $\mathbf{K}_{\eta 2} \succ \mathbf{0}$ is a diagonal gain matrix.

Using the rotational RBF estimate $\hat{\mathbf{f}}_R(\boldsymbol{\xi})$ and the ESO disturbance estimate $\hat{\mathbf{z}}_{\eta 3}$ defined in the previous subsections, the final desired moment is designed as

$$\mathbf{M}_d = \mathbf{M}_{d,bs} - \mathbf{J}_L\hat{\mathbf{f}}_R(\boldsymbol{\xi}) - \mathbf{J}_L\hat{\mathbf{z}}_{\eta 3}. \tag{35}$$

Stability Analysis

To verify the stability of the proposed composite controller under the ideal payload-layer model, this subsection analyzes the ESO observation error system and the closed-loop payload tracking error system based on Lyapunov theory. The following assumptions are introduced.

Assumption 1. The desired payload position, desired payload attitude, and their first- and second-order derivatives are continuous and bounded.

Assumption 2. For both the translational and rotational channels, the unknown nonlinear terms can be approximated by optimal constant RBF weights with bounded approximation errors.

Assumption 3. The hidden-layer outputs of the RBF neural networks are bounded.

Assumption 4. The augmented disturbance in the translational channel and the lumped disturbance in the rotational channel are continuously differentiable, and their time derivatives are bounded.

Assumption 5. The payload attitude trajectory remains in a compact subset where the local attitude-coordinate representation used for the rotational backstepping design is nonsingular.

To eliminate the effect of the known gravity term on the translational ESO error dynamics, the translational-channel augmented disturbance is redefined accordingly. Then, the translational ESO can be written in a standard augmented form.

Boundedness of ESO Observation Errors Define the translational-channel ESO estimation error as

$$\tilde{\mathbf{z}}_{o,p} = \begin{bmatrix} \tilde{\mathbf{z}}_{p1}^T & \tilde{\mathbf{z}}_{p2}^T & \tilde{\mathbf{z}}_{p3}^T \end{bmatrix}^T, \tag{36}$$

where $\tilde{\mathbf{z}}_{pj} = \mathbf{z}_{pj} - \hat{\mathbf{z}}_{pj}$, $j = 1, 2, 3$, denote the estimation errors of position, velocity, and lumped disturbance, respectively. By subtracting the observer equations from the plant equations, the translational ESO error dynamics can be written as

$$\dot{\tilde{\mathbf{z}}}_{o,p} = \mathbf{A}_p \tilde{\mathbf{z}}_{o,p} + \mathbf{B}_p \dot{\mathbf{d}}_p, \tag{37}$$

where

$$\mathbf{A}_p = \begin{bmatrix} -\beta_{p1} I_3 & I_3 & 0_{3\times3} \\ -\beta_{p2} I_3 & 0_{3\times3} & I_3 \\ -\beta_{p3} I_3 & 0_{3\times3} & 0_{3\times3} \end{bmatrix}, \qquad \mathbf{B}_p = \begin{bmatrix} 0_{3\times3} \\ 0_{3\times3} \\ I_3 \end{bmatrix}. \tag{38}$$

By standard bandwidth parameterization of the ESO gains Gao (2003), the observer-error matrix $\mathbf{A}_p$ is Hurwitz. Therefore, for any symmetric positive definite matrix $\mathbf{Q}_p$, there exists a unique symmetric positive definite matrix $\mathbf{P}_p$ satisfying

$$\mathbf{A}_p^T \mathbf{P}_p + \mathbf{P}_p \mathbf{A}_p = -\mathbf{Q}_p. \tag{39}$$

Choose the Lyapunov function

$$V_{ESO,p} = \tilde{\mathbf{z}}_{o,p}^T \mathbf{P}_p \tilde{\mathbf{z}}_{o,p}. \tag{40}$$

Differentiating $V_{ESO,p}$ along the observer error dynamics yields

$$\dot{V}_{ESO,p} = -\tilde{\mathbf{z}}_{o,p}^T \mathbf{Q}_p \tilde{\mathbf{z}}_{o,p} + 2\tilde{\mathbf{z}}_{o,p}^T \mathbf{P}_p \mathbf{B}_p \dot{\mathbf{d}}_p. \tag{41}$$

By Young's inequality and Assumption 4, one obtains

$$\dot{V}_{ESO,p} \leq -c_{p1} \|\tilde{\mathbf{z}}_{o,p}\|^2 + c_{p2}, \tag{42}$$

where $c_{p1} > 0$ and $c_{p2} > 0$ are constants. Hence, the translational ESO estimation error is uniformly ultimately bounded. Moreover, if $\dot{\mathbf{d}}_p \to 0$, then the estimation error converges to the origin exponentially.

Similarly, define the rotational-channel ESO estimation error as

$$\tilde{\mathbf{z}}_{o,\eta} = \begin{bmatrix} \tilde{\mathbf{z}}_{\eta1}^T & \tilde{\mathbf{z}}_{\eta2}^T & \tilde{\mathbf{z}}_{\eta3}^T \end{bmatrix}^T. \tag{43}$$

where $\tilde{\mathbf{z}}_{\eta j} = \mathbf{z}_{\eta j} - \hat{\mathbf{z}}_{\eta j}$, $j = 1, 2, 3$, denote the attitude, angular-rate, and lumped-disturbance estimation errors, respectively. Its error dynamics can be expressed as

$$\dot{\tilde{\mathbf{z}}}_{o,\eta} = \mathbf{A}_\eta \tilde{\mathbf{z}}_{o,\eta} + \mathbf{B}_\eta \dot{\mathbf{d}}_\eta, \tag{44}$$

where $\mathbf{A}_\eta$ is also Hurwitz due to the bandwidth-based gain selection. Thus, for any symmetric positive definite matrix $\mathbf{Q}_\eta$, there exists a unique symmetric positive definite matrix $\mathbf{P}_\eta$ such that

$$\mathbf{A}_\eta^T \mathbf{P}_\eta + \mathbf{P}_\eta \mathbf{A}_\eta = -\mathbf{Q}_\eta. \tag{45}$$

Choose

$$V_{ESO,\eta} = \tilde{\mathbf{z}}_{o,\eta}^T \mathbf{P}_\eta \tilde{\mathbf{z}}_{o,\eta}. \tag{46}$$

Following the same procedure gives

$$\dot{V}_{ESO,\eta} \leq -c_{\eta 1} \|\tilde{\mathbf{z}}_{o,\eta}\|^2 + c_{\eta 2}, \tag{47}$$

where $c_{\eta 1} > 0$ and $c_{\eta 2} > 0$ are constants. Therefore, the rotational ESO estimation error is uniformly ultimately bounded; when $\dot{\mathbf{d}}_\eta \to 0$, it converges exponentially to the origin.

Closed-Loop Stability of the Composite Controller After establishing the boundedness of the ESO estimation errors, the overall closed-loop error system of the RBF-adaptive Backstepping-ESO controller is analyzed. Let the unknown acceleration-level translational uncertainty and rotational uncertainty be written as

$$\mathbf{f}_p^*(\boldsymbol{\xi}) = \boldsymbol{\Theta}_p^{*T} \bar{\mathbf{h}}(\boldsymbol{\xi}) + \boldsymbol{\varepsilon}_p^r, \qquad \mathbf{f}_\eta^*(\boldsymbol{\xi}) = \boldsymbol{\Theta}_R^{*T} \bar{\mathbf{h}}(\boldsymbol{\xi}) + \boldsymbol{\varepsilon}_\eta^r, \tag{48}$$

where $\boldsymbol{\Theta}_p^*$ and $\boldsymbol{\Theta}_R^*$ are stacked ideal constant RBF weight matrices consistent with the channel-wise vectors $\boldsymbol{\theta}_{p,i}$ and $\boldsymbol{\theta}_{R,i}$ defined above, and $\boldsymbol{\varepsilon}_p^r$ and $\boldsymbol{\varepsilon}_\eta^r$ are bounded approximation errors. The implemented stacked estimates are

$$\hat{\boldsymbol{\Theta}}_p = \begin{bmatrix} \hat{\boldsymbol{\theta}}_{p,1} & \hat{\boldsymbol{\theta}}_{p,2} & \hat{\boldsymbol{\theta}}_{p,3} \end{bmatrix}, \qquad \hat{\boldsymbol{\Theta}}_R = \begin{bmatrix} \hat{\boldsymbol{\theta}}_{R,1} & \hat{\boldsymbol{\theta}}_{R,2} & \hat{\boldsymbol{\theta}}_{R,3} \end{bmatrix}, \tag{49}$$

so that $\hat{\mathbf{f}}_p = \hat{\boldsymbol{\Theta}}_p^T \bar{\mathbf{h}}$ and $\hat{\mathbf{f}}_R = \hat{\boldsymbol{\Theta}}_R^T \bar{\mathbf{h}}$. Define the RBF weight estimation errors and ESO disturbance-estimation errors as

$$\tilde{\boldsymbol{\Theta}}_p = \hat{\boldsymbol{\Theta}}_p - \boldsymbol{\Theta}_p^*, \qquad \tilde{\boldsymbol{\Theta}}_R = \hat{\boldsymbol{\Theta}}_R - \boldsymbol{\Theta}_R^*, \tag{50}$$

$$\tilde{\mathbf{z}}_{p3} = \mathbf{z}_{p3} - \hat{\mathbf{z}}_{p3}, \qquad \tilde{\mathbf{z}}_{\eta 3} = \mathbf{z}_{\eta 3} - \hat{\mathbf{z}}_{\eta 3}. \tag{51}$$

Then the residual terms that cannot be cancelled by the implemented compensation are

$$\boldsymbol{\Delta}_p = \boldsymbol{\varepsilon}_p^r + \tilde{\mathbf{z}}_{p3}, \qquad \boldsymbol{\Delta}_\eta = \boldsymbol{\varepsilon}_\eta^r + \tilde{\mathbf{z}}_{\eta 3}. \tag{52}$$

According to Assumption 2 and the ESO boundedness result, there exist constants $\bar{\Delta}_p, \bar{\Delta}_\eta > 0$ such that $\|\boldsymbol{\Delta}_p\| \le \bar{\Delta}_p$ and $\|\boldsymbol{\Delta}_\eta\| \le \bar{\Delta}_\eta$.

Substituting the proposed control law into the translational and rotational dynamics yields the closed-loop error systems for the two channels. Define the composite tracking vectors

$$\boldsymbol{\chi}_p = \begin{bmatrix} \mathbf{e}_{p_L}^T & \mathbf{z}_p^T \end{bmatrix}^T, \qquad \boldsymbol{\chi}_\eta = \begin{bmatrix} \mathbf{e}_{\eta_L}^T & \mathbf{z}_\eta^T \end{bmatrix}^T. \tag{53}$$

To interpret the channel-wise adaptive law (16), note that the same six-dimensional vector $\boldsymbol{\xi}$ is used in both the RBF approximation and the weight update. The scalar factor $\boldsymbol{\xi}^T \mathbf{P}_i \mathbf{B}_i$ is the i-th channel projection of the Lyapunov cross term and does not define a new input variable. Using (27)–(35), the translational and rotational error dynamics can be expressed in the compact form

$$\dot{\boldsymbol{\chi}}_p = \mathbf{A}_{c,p}\boldsymbol{\chi}_p + \mathbf{B}_{c,p}\left(-\tilde{\boldsymbol{\Theta}}_p^T \bar{\mathbf{h}}(\boldsymbol{\xi}) + \boldsymbol{\Delta}_p\right), \tag{54}$$

$$\dot{\boldsymbol{\chi}}_\eta = \mathbf{A}_{c,\eta}\boldsymbol{\chi}_\eta + \mathbf{B}_{c,\eta}\left(-\tilde{\boldsymbol{\Theta}}_R^T \bar{\mathbf{h}}(\boldsymbol{\xi}) + \boldsymbol{\Delta}_\eta\right), \tag{55}$$

where

$$\mathbf{A}_{c,p} = \begin{bmatrix} -\mathbf{K}_{p1} & \mathbf{I}_3 \\ \mathbf{0}_{3\times 3} & -\mathbf{K}_{p2} \end{bmatrix}, \quad \mathbf{A}_{c,\eta} = \begin{bmatrix} -\mathbf{K}_{\eta 1} & \mathbf{I}_3 \\ \mathbf{0}_{3\times 3} & -\mathbf{K}_{\eta 2} \end{bmatrix}, \tag{56}$$

$$\mathbf{B}_{c,p} = \mathbf{B}_{c,\eta} = \begin{bmatrix} \mathbf{0}_{3\times 3} \\ \mathbf{I}_3 \end{bmatrix}. \tag{57}$$

Because $\mathbf{K}_{p1}$, $\mathbf{K}_{p2}$, $\mathbf{K}_{\eta 1}$, and $\mathbf{K}_{\eta 2}$ are positive definite diagonal matrices, both $\mathbf{A}_{c,p}$ and $\mathbf{A}_{c,\eta}$ are block upper-triangular Hurwitz matrices. Therefore, for arbitrary symmetric positive definite matrices $\mathbf{Q}_1$ and $\mathbf{Q}_2$, there exist unique symmetric positive definite matrices $\mathbf{P}_1$ and $\mathbf{P}_2$ satisfying

$$\mathbf{A}_{c,p}^T \mathbf{P}_1 + \mathbf{P}_1 \mathbf{A}_{c,p} = -\mathbf{Q}_1, \qquad \mathbf{A}_{c,\eta}^T \mathbf{P}_2 + \mathbf{P}_2 \mathbf{A}_{c,\eta} = -\mathbf{Q}_2. \tag{58}$$

Choose the Lyapunov candidate function as

$$V = \frac{1}{2}\boldsymbol{\chi}_p^T\mathbf{P}_1\boldsymbol{\chi}_p + \frac{1}{2}\boldsymbol{\chi}_\eta^T\mathbf{P}_2\boldsymbol{\chi}_\eta + \frac{1}{2}\sum_{i=1}^{3}\tilde{\boldsymbol{\theta}}_{p,i}^T\boldsymbol{\Gamma}_{p,i}^{-1}\tilde{\boldsymbol{\theta}}_{p,i} + \frac{1}{2}\sum_{i=1}^{3}\tilde{\boldsymbol{\theta}}_{R,i}^T\boldsymbol{\Gamma}_{R,i}^{-1}\tilde{\boldsymbol{\theta}}_{R,i}, \tag{59}$$

where $\tilde{\boldsymbol{\theta}}_{p,i}$ and $\tilde{\boldsymbol{\theta}}_{R,i}$ are the channel-wise RBF weight estimation errors, and $\boldsymbol{\Gamma}_{p,i} = \gamma_{p,i}\mathbf{I}_5$ and $\boldsymbol{\Gamma}_{R,i} = \gamma_{R,i}\mathbf{I}_5$ are the nominal positive definite channel-wise adaptive gain matrices.

Since all weighting matrices are positive definite, V is positive definite and radially unbounded with respect to the composite tracking errors and RBF weight estimation errors. Differentiating V along (54) and (55) gives

$$\begin{aligned}\dot{V} = & -\frac{1}{2}\boldsymbol{\chi}_p^T\mathbf{Q}_1\boldsymbol{\chi}_p - \frac{1}{2}\boldsymbol{\chi}_\eta^T\mathbf{Q}_2\boldsymbol{\chi}_\eta \\ & + \boldsymbol{\chi}_p^T\mathbf{P}_1\mathbf{B}_{c,p}\left(-\tilde{\boldsymbol{\Theta}}_p^T\bar{\mathbf{h}} + \boldsymbol{\Delta}_p\right) \\ & + \boldsymbol{\chi}_\eta^T\mathbf{P}_2\mathbf{B}_{c,\eta}\left(-\tilde{\boldsymbol{\Theta}}_R^T\bar{\mathbf{h}} + \boldsymbol{\Delta}_\eta\right) \\ & + \sum_{i=1}^{3}\tilde{\boldsymbol{\theta}}_{p,i}^T\boldsymbol{\Gamma}_{p,i}^{-1}\dot{\hat{\boldsymbol{\theta}}}_{p,i} + \sum_{i=1}^{3}\tilde{\boldsymbol{\theta}}_{R,i}^T\boldsymbol{\Gamma}_{R,i}^{-1}\dot{\hat{\boldsymbol{\theta}}}_{R,i}.\end{aligned} \tag{60}$$

The nominal part of the Lyapunov-based adaptive law (16) cancels the RBF cross terms in (60). When the dead-zone scaling makes $\gamma_{i,\mathrm{eff}} < \gamma_i$, the corresponding uncancelled term is active only in the bounded small-error region and is absorbed into the residual constant below. The leakage term introduces negative quadratic damping for the estimated weights and a bounded constant depending on the ideal weights. Specifically, by Young's inequality, the leakage contribution satisfies an inequality of the form

$$\tilde{\boldsymbol{\theta}}_i^T\boldsymbol{\Gamma}_i^{-1}\left(-\kappa\boldsymbol{\Gamma}_i\hat{\boldsymbol{\theta}}_i\right) \le -\frac{\kappa}{2}\|\tilde{\boldsymbol{\theta}}_i\|^2 + \frac{\kappa}{2}\|\boldsymbol{\theta}_i^*\|^2. \tag{61}$$

The residual terms are also bounded by Young's inequality:

$$\begin{aligned}\boldsymbol{\chi}_p^T\mathbf{P}_1\mathbf{B}_{c,p}\boldsymbol{\Delta}_p &\le \delta_p\|\boldsymbol{\chi}_p\|^2 + \frac{\|\mathbf{P}_1\mathbf{B}_{c,p}\|^2\bar{\Delta}_p^2}{4\delta_p}, \\ \boldsymbol{\chi}_\eta^T\mathbf{P}_2\mathbf{B}_{c,\eta}\boldsymbol{\Delta}_\eta &\le \delta_\eta\|\boldsymbol{\chi}_\eta\|^2 + \frac{\|\mathbf{P}_2\mathbf{B}_{c,\eta}\|^2\bar{\Delta}_\eta^2}{4\delta_\eta},\end{aligned} \tag{62}$$

where $\delta_p > 0$ and $\delta_\eta > 0$ can be selected such that $\lambda_{\min}(\mathbf{Q}_1)/2 - \delta_p > 0$ and $\lambda_{\min}(\mathbf{Q}_2)/2 - \delta_\eta > 0$. Therefore,

$$\begin{aligned}\dot{V} \leq & -\lambda_1\|\boldsymbol{\chi}_p\|^2 - \lambda_2\|\boldsymbol{\chi}_\eta\|^2 - \lambda_3\|\tilde{\boldsymbol{\Theta}}_p\|_F^2 \\ & - \lambda_4\|\tilde{\boldsymbol{\Theta}}_R\|_F^2 + c,\end{aligned} \tag{63}$$

where $\lambda_1, \lambda_2, \lambda_3, \lambda_4 > 0$ are constants, and $c > 0$ collects the bounded RBF approximation residuals, bounded ESO estimation residuals, and bounded leakage terms involving the ideal weights. Since V is quadratically bounded by the error vector

$$\mathbf{y} = \begin{bmatrix}\boldsymbol{\chi}_p^T & \boldsymbol{\chi}_\eta^T & \mathrm{vec}(\tilde{\boldsymbol{\Theta}}_p)^T & \mathrm{vec}(\tilde{\boldsymbol{\Theta}}_R)^T\end{bmatrix}^T, \tag{64}$$

there exist positive constants $\underline{\alpha}_V$ and $\overline{\alpha}_V$ such that

$$\underline{\alpha}_V\|\mathbf{y}\|^2 \leq V \leq \overline{\alpha}_V\|\mathbf{y}\|^2. \tag{65}$$

Consequently, there exist positive constants α and β such that

$$\dot{V} \leq -\alpha V + \beta. \tag{66}$$

Solving (66) gives

$$V(t) \leq V(0)e^{-\alpha t} + \frac{\beta}{\alpha}\left(1 - e^{-\alpha t}\right). \tag{67}$$

Thus, $V(t)$ is bounded for all $t \geq 0$, and all components of $\mathbf{y}$ are uniformly ultimately bounded with ultimate radius proportional to $\sqrt{\beta/\alpha}$. According to the uniform ultimate boundedness criterion of Lyapunov theory, under Assumptions 1–5, the position tracking error, attitude tracking error, velocity-level backstepping errors, and RBF parameter estimation errors of the closed-loop payload-layer system are uniformly ultimately bounded.

Because the desired trajectories are bounded by Assumption 1, boundedness of $\mathbf{e}_{p_L}$, $\mathbf{z}_p$, $\mathbf{e}_{\eta_L}$, and $\mathbf{z}_\eta$ also implies bounded payload velocity and attitude-rate tracking errors. Together with the bounded ESO errors and projected/leakage-stabilized RBF weights, the compensation terms in (30) and (35) remain bounded. Hence, the proposed composite controller guarantees boundedness of the ideal payload-layer closed-loop signals and achieves bounded tracking of the payload position and attitude. If the RBF approximation residuals and ESO disturbance-estimation residuals converge to zero, then β can be made zero in the ideal case and the tracking errors converge asymptotically to the origin.

Remark The above analysis shows that the Backstepping terms establish the nominal error convergence structure, the RBF neural networks compensate for the major unknown nonlinearities, and the ESO further estimates and suppresses the remaining lumped disturbances online. As a result, the proposed composite controller maintains the closed-loop stability and robustness of the payload transportation system under bounded uncertainties and external disturbances.

Control Allocation and Cable Force Generation

Following the geometric cooperative transport formulation in Lee et al. (2013a), the payload-level controller first computes the desired net wrench for payload motion tracking, consisting of a resultant force $\mathbf{F}_d \in \mathbb{R}^3$ expressed in the inertial frame and a resultant moment $\mathbf{M}_d \in \mathbb{R}^3$ expressed in the payload body frame. This pair $(\mathbf{F}_d, \mathbf{M}_d)$ provides a compact interface between high-level tracking control and low-level actuation, because it captures both translational and rotational objectives in a unified form. Since the payload is driven through multiple cables rather than a single actuator, the control-allocation module then maps the desired wrench to individual cable forces while preserving the overall force–moment balance required by the payload dynamics.

The corresponding desired resultant force and moment acting on the payload are given by

$$\mathbf{F}_d = \sum_{i=1}^{n} \mathbf{T}_{i,d}, \qquad \mathbf{M}_d = \sum_{i=1}^{n} \hat{\boldsymbol{\rho}}_i \mathbf{R}_L^T \mathbf{T}_{i,d}. \tag{68}$$

where $\mathbf{T}_{i,d}$ denotes the desired tension vector exerted by the i-th cable on the payload. This notation is consistent with the payload dynamics, where $\mathbf{T}_i = -\mu_i \mathbf{q}_i$ and $\mu_i > 0$ is the scalar tension magnitude.

Let $\boldsymbol{\rho}_i \in \mathbb{R}^3$ denote the attachment point of the i-th cable expressed in the payload body frame, and define the configuration matrix

$$\mathbf{P} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{I}_3 & \cdots & \mathbf{I}_3 \\ \hat{\boldsymbol{\rho}}_1 & \hat{\boldsymbol{\rho}}_2 & \cdots & \hat{\boldsymbol{\rho}}_n \end{bmatrix}, \qquad \hat{\boldsymbol{\rho}}_i = S(\boldsymbol{\rho}_i). \tag{69}$$

Since the desired force is computed in the inertial frame while the desired moment is computed in the payload body frame, the commanded wrench is written as

$$\mathbf{w}_d = \begin{bmatrix} \mathbf{R}_L^T \mathbf{F}_d \\ \mathbf{M}_d \end{bmatrix}. \tag{70}$$

The minimum-norm cable tensions in body coordinates are then obtained by

$$\mathbf{T}_d^b = \mathbf{P}^T(\mathbf{P}\mathbf{P}^T)^{-1}\mathbf{w}_d, \tag{71}$$

and the Moore–Penrose pseudoinverse is used when $\mathbf{P}\mathbf{P}^T$ is singular. The desired inertial-frame cable tensions are recovered by rotating each block back to the world frame:

$$\mathbf{T}_d = diag(\mathbf{R}_L, \ldots, \mathbf{R}_L)\mathbf{T}_d^b, \tag{72}$$

where

$$\mathbf{T}_d = \begin{bmatrix} \mathbf{T}_{1,d} \\ \vdots \\ \mathbf{T}_{n,d} \end{bmatrix}. \tag{73}$$

Because each cable can only transmit force along its current direction $\mathbf{q}_i$, the realized cable force is taken as the projection

$$\mathbf{T}_i = \mathbf{q}_i\mathbf{q}_i^T\mathbf{T}_{i,d}. \tag{74}$$

The corresponding resultant payload force and moment are

$$\mathbf{F}_{in} = \sum_{i=1}^{n}\mathbf{T}_i, \qquad \mathbf{M}_{in} = \sum_{i=1}^{n}\hat{\boldsymbol{\rho}}_i\mathbf{R}_L^T\mathbf{T}_i. \tag{75}$$

These quantities are used in the simulation to evaluate the allocation error between the commanded wrench $(\mathbf{F}_d, \mathbf{M}_d)$ and the actually generated wrench $(\mathbf{F}_{in}, \mathbf{M}_{in})$.

Cable Direction Tracking and Quadrotor Force Command

Following Lee et al. (2013a), after allocation the desired cable direction for the i-th UAV is defined from the allocated tension as

$$\mathbf{q}_{i,d} = -\frac{\mathbf{T}_{i,d}}{\|\mathbf{T}_{i,d}\|}. \tag{76}$$

and the desired cable angular velocity is computed as

$$\boldsymbol{\omega}_{i,d} = \mathbf{q}_{i,d} \times \dot{\mathbf{q}}_{i,d}. \tag{77}$$

The cable tracking errors are chosen as

$$\mathbf{e}_{q,i} = \mathbf{q}_{i,d} \times \mathbf{q}_i, \qquad \mathbf{e}_{\omega,i} = \boldsymbol{\omega}_i + \hat{\mathbf{q}}_i^2\boldsymbol{\omega}_{i,d}. \tag{78}$$

To account for payload translation and rotation, the feedforward term for the i-th cable is

$$\mathbf{a}_i = \frac{1}{m_L}\mathbf{F}_{in} + \mathbf{R}_L\hat{\boldsymbol{\Omega}}_L^2\boldsymbol{\rho}_i + \mathbf{R}_L\hat{\boldsymbol{\rho}}_i\mathbf{J}_L^{-1}\left(\hat{\boldsymbol{\Omega}}_L\mathbf{J}_L\boldsymbol{\Omega}_L - \mathbf{M}_{in}\right). \tag{79}$$

The total force command generated for the i-th quadrotor is decomposed into a normal component and a parallel component,

$$\mathbf{u}_i = \mathbf{u}_{i,\perp} + \mathbf{u}_{i,\parallel}. \tag{80}$$

The normal component is

$$\begin{aligned}\mathbf{u}_{i,\perp} = {} & m_{p,i}l_i\hat{\mathbf{q}}_i\left(-K_{q,i}\mathbf{e}_{q,i} - K_{\omega,i}\mathbf{e}_{\omega,i} - (\mathbf{q}_i^T\boldsymbol{\omega}_{i,d})\dot{\mathbf{q}}_i - \hat{\mathbf{q}}_i^2\dot{\boldsymbol{\omega}}_{i,d}\right) \\ & - m_{p,i}\hat{\mathbf{q}}_i^2\mathbf{a}_i,\end{aligned} \tag{81}$$

while the parallel component is

$$\mathbf{u}_{i,\parallel} = \mathbf{T}_i + m_{p,i}\left(l_i\|\boldsymbol{\omega}_i\|^2\mathbf{q}_i + \mathbf{q}_i\mathbf{q}_i^T\mathbf{a}_i\right). \tag{82}$$

Hence, each UAV receives a commanded inertial-force vector $\mathbf{u}_i$ from the allocation module.

Quadrotor Thrust and Attitude Command

Following the geometric thrust-direction construction in Lee et al. (2013a,b), the cable-level controller provides for each quadrotor a commanded force vector $\mathbf{u}_i \in \mathbb{R}^3$ in the inertial frame. The role of the inner-loop UAV controller is to convert this vector into a thrust command and a corresponding attitude reference.

The desired body-fixed thrust direction is chosen to align with the commanded force, namely,

$$\mathbf{b}_{3,i,d} = \frac{\mathbf{u}_i}{\|\mathbf{u}_i\|}. \tag{83}$$

To complete the attitude specification, a desired yaw angle $\psi_{i,d}$ is introduced through the horizontal reference direction

$$\mathbf{b}_{1,\psi,i} = \begin{bmatrix}\cos\psi_{i,d} \\ \sin\psi_{i,d} \\ 0\end{bmatrix}. \tag{84}$$

An orthonormal desired body frame is then constructed as

$$\mathbf{b}_{2,i,d} = \frac{\mathbf{b}_{3,i,d} \times \mathbf{b}_{1,\psi,i}}{\|\mathbf{b}_{3,i,d} \times \mathbf{b}_{1,\psi,i}\|}, \qquad \mathbf{b}_{1,i,d} = \mathbf{b}_{2,i,d} \times \mathbf{b}_{3,i,d}, \tag{85}$$

which yields the desired rotation matrix

$$\mathbf{R}_{i,d} = \begin{bmatrix} \mathbf{b}_{1,i,d} & \mathbf{b}_{2,i,d} & \mathbf{b}_{3,i,d} \end{bmatrix}. \tag{86}$$

The thrust magnitude applied by the i-th quadrotor is obtained by projecting the commanded force onto the current body z-axis,

$$f_i = \mathbf{u}_i^T \mathbf{R}_i \mathbf{e}_3, \tag{87}$$

where $\mathbf{R}_i \in \mathsf{SO}(3)$ denotes the current attitude matrix and $\mathbf{e}_3 = [0, 0, 1]^T$.

Although the desired attitude is generated geometrically, the attitude tracking controller used in this work is implemented in Euler-angle coordinates. Let

$$\boldsymbol{\eta}_i = [\phi_i, \theta_i, \psi_i]^T, \qquad \boldsymbol{\eta}_{i,d} = [\phi_{i,d}, \theta_{i,d}, \psi_{i,d}]^T \tag{88}$$

denote the current and desired Euler-angle vectors, respectively, with $\boldsymbol{\eta}_{i,d}$ extracted from $\mathbf{R}_{i,d}$. The attitude tracking errors are defined as

$$\mathbf{z}_{1,i} = \boldsymbol{\eta}_i - \boldsymbol{\eta}_{i,d}, \qquad \mathbf{z}_{2,i} = \dot{\boldsymbol{\eta}}_i - \dot{\boldsymbol{\eta}}_{i,d} + \mathbf{C}_{1,i}\mathbf{z}_{1,i}, \tag{89}$$

where $\mathbf{C}_{1,i}$ is a positive definite gain matrix. The control moment is then designed as

$$\boldsymbol{\tau}_i = \mathbf{B}_i^{-1} \left(-\mathbf{z}_{1,i} - \mathbf{c}_i(\dot{\boldsymbol{\eta}}_i) + \dot{\boldsymbol{\alpha}}_{1,i} - \mathbf{C}_{2,i}\mathbf{z}_{2,i} \right), \tag{90}$$

where $\mathbf{B}_i$ is the input gain matrix associated with the quadrotor attitude dynamics, $\mathbf{C}_{2,i}$ is a second positive definite gain matrix, and $\mathbf{c}_i(\dot{\boldsymbol{\eta}}_i)$ represents the Coriolis/gyroscopic coupling term of the rotational subsystem.

Consequently, the force command $\mathbf{u}_i$ obtained from the cable-load coordination layer is transformed into realizable thrust and moment commands for each quadrotor through the desired thrust direction, yaw specification, and Euler-angle-based inner-loop controller.

Simulation Setup

A high-fidelity numerical simulation environment was developed in MATLAB to assess the tracking performance and disturbance rejection capability of the proposed cooperative load-lifting system. The simulated platform consists of three identical quadrotors transporting a rigid rectangular payload through three taut flexible cables with fixed nominal length. The overall simulation is implemented as a hierarchical closed-loop system comprising payload-level control, force allocation, cable-direction regulation, and low-level quadrotor attitude control.

Physical Parameters and Numerical Integration

The coupled payload–quadrotor dynamics are integrated using a fixed-step fourth-order Runge–Kutta (RK4) scheme. A sampling interval of $\Delta t = 0.001$ s is adopted to capture the fast attitude and cable-direction dynamics, and each simulation runs for a total duration of $T_f = 20$ s.

The principal numerical, physical, and geometric parameters used in the simulations are listed in the Appendix. Unless otherwise stated, the three quadrotors are identical, each with mass $m_p = 2.0$ kg and inertia matrix $J_p = \mathrm{diag}(0.082,\ 0.0845,\ 0.1377)\ \mathrm{kg}\cdot\mathrm{m}^2$. The payload mass is $m_L = 2.0$ kg, its geometric dimensions are $1.0 \times 0.8 \times 0.2$ m, and its inertia matrix is $J_L = \mathrm{diag}(0.1133,\ 0.1733,\ 0.2733)\ \mathrm{kg}\cdot\mathrm{m}^2$.

Initial Conditions and Geometric Configuration

The quadrotors are initialized from a near-hover configuration with zero attitude angles, zero angular rates, and zero translational velocities. Their initial positions are

$$\begin{aligned} \mathbf{p}_1(0) &= [0.5,\ 4.2,\ 0]^T, \\ \mathbf{p}_2(0) &= [-0.5,\ 4.6,\ 0]^T, \\ \mathbf{p}_3(0) &= [-0.5,\ 3.8,\ 0]^T. \end{aligned} \tag{91}$$

This slightly asymmetric arrangement avoids perfectly symmetric transient behavior at the start of the simulation.

The payload is initially placed at

$$\mathbf{p}_L(0) = [0,\ 4.2,\ 0.1]^T,$$

with zero translational velocity, zero angular displacement, and zero angular velocity. The attachment-point vectors expressed in the payload

body frame are defined as

$$\begin{aligned}\boldsymbol{\rho}_1 &= [0.5,\ 0.0,\ 0.1]^T,\\ \boldsymbol{\rho}_2 &= [-0.5,\ 0.4,\ 0.1]^T,\\ \boldsymbol{\rho}_3 &= [-0.5,\ -0.4,\ 0.1]^T.\end{aligned} \tag{92}$$

Operational Phases and Reference Trajectory

The simulation is divided into two sequential phases.

First, during the takeoff phase, each quadrotor ascends vertically to a prescribed hovering position above its corresponding payload attachment point. The target altitude is selected as

$$z_i^\star = l_i + h_L = 1.0 + 0.2 = 1.2\ \mathrm{m},$$

where h_L denotes the payload height. The takeoff phase is terminated once all quadrotors satisfy the altitude tolerance criterion of 10^{-3} m.

Second, after takeoff is completed, the payload tracking phase begins. At this point, the payload reference trajectory is generated with its time origin reset to zero, so that the commanded transport task is not time-shifted by the takeoff duration. The desired payload position is given by

$$\mathbf{p}_{L,d}(t) = \begin{bmatrix} 1.5\sin(\omega_d t) \\ 4.2\cos(\omega_d t/2) \\ 0.5 \end{bmatrix}, \qquad \omega_d = \frac{2\pi}{10}\ \mathrm{rad/s}, \tag{93}$$

with analytically generated velocity and acceleration references.

The desired payload attitude is constructed from the tangent direction of the reference trajectory. Specifically, for $\|\dot{\mathbf{p}}_{L,d}\| \geq 10^{-3}$,

$$\mathbf{r}_{d1} = \frac{\dot{\mathbf{p}}_{L,d}}{\|\dot{\mathbf{p}}_{L,d}\|}, \qquad \mathbf{r}_{d3} = \mathbf{e}_3, \qquad \mathbf{r}_{d2} = \frac{\mathbf{r}_{d3} \times \mathbf{r}_{d1}}{\|\mathbf{r}_{d3} \times \mathbf{r}_{d1}\|}, \tag{94}$$

and the desired rotation matrix is defined as

$$R_{L,d} = \begin{bmatrix} \mathbf{r}_{d1} & \mathbf{r}_{d2} & \mathbf{r}_{d3} \end{bmatrix}. \tag{95}$$

When the desired translational speed is close to zero, the reference attitude is set to $R_{L,d} = I_3$. This construction aligns the payload heading with the instantaneous direction of motion while maintaining a horizontal body z-axis.

Control Architecture and Disturbance Injection

To avoid repeating the previously introduced control pipeline, this subsection focuses only on the disturbance setting used for robustness evaluation. When disturbance injection is enabled, the translational and rotational disturbance terms are modeled as

$$d_p(t) = 3\sin t + 0.2\sin 2t + 0.3\sin 3t + 0.4\sin 4t, \tag{96}$$

$$d_R(t) = 1\sin t + 0.2\sin 2t + 0.3\sin 3t + 0.4\sin 4t. \tag{97}$$

In the implemented model, the same scalar disturbance $d_p(t)$ is injected into each translational channel of the payload dynamics, while $d_R(t)$ is injected into each rotational channel.

Evaluation Protocol

The simulation study is organized as a backstepping-family ablation so that the effect of each compensation module can be evaluated under the same nominal controller structure. Four payload-level controllers are evaluated under both no-disturbance and disturbance-injection conditions:

- **Backstepping**, the nominal recursive nonlinear tracking controller;
- **RBF+Backstepping**, where the RBF neural network is added for online approximation of unmodeled nonlinear dynamics;
- **ESO+Backstepping**, where the ESO is added for online estimation of lumped residual disturbances;
- **RBF+ESO+Backstepping**, the proposed composite controller combining neural approximation and observer-based disturbance compensation.

All controllers are evaluated under the same physical parameters, initial conditions, reference trajectory, and disturbance model. The reported RMS indices are computed from payload position and attitude errors over the final 80% of the stored simulation data. The controller and observer parameters used in the ablation runs are listed separately in the Appendix. The standalone Backstepping parameter set is used by the pure Backstepping case, while the RBF- and ESO-enabled cases share a common module-toggle baseline and differ only by the switches specified in Table 1.

Table 1. Backstepping-family ablation cases

Controller	RBF compensation	ESO compensation
Backstepping	Off	Off
Backstepping+RBF	On	Off
Backstepping+ESO	Off	On
Backstepping+RBF+ESO	On	On

Simulation Results and Discussion

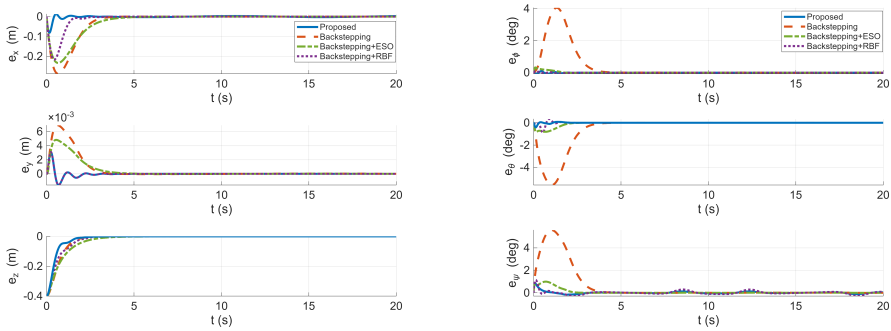
This section follows the ablation protocol defined in Section . The proposed method is decomposed into its constituent modules and compared with plain backstepping under nominal and disturbed conditions.

Backstepping-Family Ablation Design

The ablation study is designed to isolate the contribution of each compensation module while keeping the nominal backstepping structure, reference trajectory, allocation module, and UAV-level controller unchanged. The four ablation cases are summarized in Table 1. Each case is evaluated twice: once without external disturbance and once with the matched multi-harmonic disturbance terms.

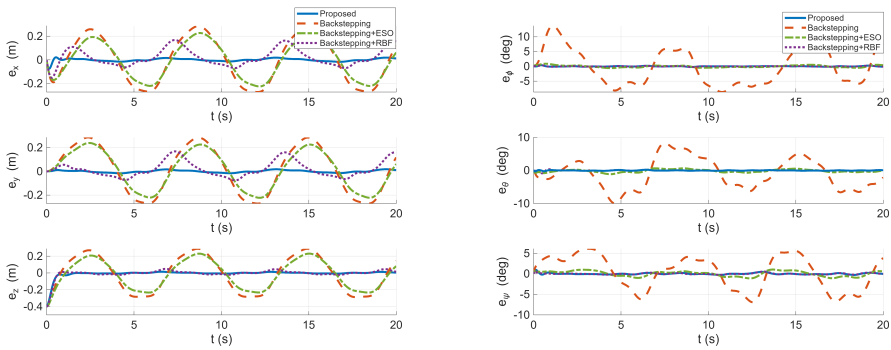
Figures 4 and 5 present the payload position and attitude responses for the backstepping-family controllers under the two disturbance settings. The time-domain curves are kept in paired figure blocks to avoid splitting the two payload-error channels across unrelated pages.

From Fig. 4, it can be observed that all backstepping-family controllers stabilize the payload tracking errors under the no-disturbance condition. However, their transient responses are clearly different. The plain Backstepping controller exhibits relatively larger attitude deviations during the initial stage, especially in the roll and pitch channels, while the controllers with RBF or ESO compensation reduce the transient oscillations to different degrees. The proposed Backstepping+RBF+ESO controller shows the fastest attenuation of the initial position and attitude errors and maintains nearly zero steady-state errors after the transient process. This indicates that even in the nominal case, the additional compensation modules help suppress the effects of nonlinear coupling and allocation-induced modeling errors.



(a) Position error (b) Attitude error

Figure 4. Backstepping-family ablation results without disturbance injection. The RBF and ESO modules improve the nominal backstepping response while preserving the same reference trajectory and allocation structure.



(a) Position error (b) Attitude error

Figure 5. Backstepping-family ablation results with disturbance injection. The complete Backstepping+RBF+ESO structure gives the most balanced reduction of payload position and attitude errors under the matched disturbance model.

Figure 5 further demonstrates the importance of disturbance compensation. When the external multi-harmonic disturbance is injected, the plain Backstepping controller shows obvious periodic tracking errors in both position and attitude channels. The RBF-only and ESO-only cases improve the response compared with plain Backstepping, but neither of them fully suppresses the disturbance-induced oscillations. In contrast, the complete Backstepping+RBF+ESO controller keeps all three position-error components within a much smaller range and also significantly

reduces the attitude-error oscillations. This result suggests that the RBF network and ESO play complementary roles: the RBF module compensates for structured nonlinear uncertainty, while the ESO estimates the remaining lumped disturbance online.

Conclusion

This paper presented an RBF-ESO backstepping control framework for multi-UAV cooperative payload transportation. The proposed controller uses backstepping to construct the nominal payload force and moment commands, an RBF neural network to approximate unmodeled nonlinear dynamics, and an ESO to estimate residual lumped disturbances online.

The simulation framework was rebuilt around a backstepping-family ablation study. Under no-disturbance and disturbance-injection conditions, the paper compares plain Backstepping, RBF+Backstepping, ESO+Backstepping, and RBF+ESO+Backstepping, which clarifies the individual contributions of the RBF and ESO modules. The results indicate that combining neural approximation with observer-based disturbance compensation improves tracking accuracy and disturbance rejection for the coupled aerial payload transportation system.

Overall, the proposed composite control structure provides a systematic way to integrate model-based backstepping, neural-network approximation, and observer-based disturbance estimation within a cooperative payload-transport framework. The ablation results show that the RBF and ESO components complement the nominal backstepping controller, yielding improved robustness without sacrificing attitude-tracking performance. These results demonstrate the effectiveness of the proposed method for accurate and robust multi-UAV payload transportation under nonlinear coupling and external disturbances.

Simulation Parameters

The numerical, physical, geometric, controller, and observer parameters used in the simulation study are listed in Tables 2 and 3.

Table 2. Physical, geometric, and numerical parameters of the cooperative load-lifting system.

Parameter	Symbol	Value
Number of quadrotors	N	3
Integration step and duration	$\Delta t,\ T_f$	$0.001\ \mathrm{s},\ 20\ \mathrm{s}$
Quadrotor mass	m_p	2.0 kg for each UAV
Quadrotor inertia	J_p	$\mathrm{diag}(0.082,\ 0.0845,\ 0.1377)\ \mathrm{kg}\cdot\mathrm{m}^2$
Payload mass	m_L	$2.0\ \mathrm{kg}$
Payload inertia	J_L	$\mathrm{diag}(0.113333,\ 0.173333,\ 0.273333)\ \mathrm{kg}\cdot\mathrm{m}^2$
Payload dimensions	$L \times W \times H$	$1.0 \times 0.8 \times 0.2\ \mathrm{m}$
Cable length	l_i	1.0 m for all three cables
Gravitational acceleration	g	$9.81\ \mathrm{m/s}^2$
Initial quadrotor positions	$\mathbf{p}_i(0)$	$[0.5, 4.2, 0]^T$, $[-0.5, 4.6, 0]^T$, $[-0.5, 3.8, 0]^T$
Initial payload position	$\mathbf{p}_L(0)$	$[0, 4.2, 0.1]^T$
Payload attachment points	$\boldsymbol{\rho}_i$	$[0.5, 0, 0.1]^T$, $[-0.5, 0.4, 0.1]^T$, $[-0.5, -0.4, 0.1]^T$

Table 3. Controller, allocation, RBF, and ESO parameters used in the ablation simulations.

Parameter group	No disturbance	With disturbance
Standalone Backstepping allocation	$K_{\Omega q} = [43.7, 28.23, 17.54]$, $K_q = [50, 50.1, 50]$, $\alpha_f = 0.30$	$K_{\Omega q} = [50, 35, 25]$, $K_q = [60, 60, 60]$, $\alpha_f = 0.25$
Standalone Backstepping payload gains	$c_1 = c_3 = \mathrm{diag}(3, 3, 3)$, $c_2 = c_4 = \mathrm{diag}(1.5, 1.5, 1.5)$	$c_1 = c_3 = \mathrm{diag}(4, 4, 4)$, $c_2 = c_4 = \mathrm{diag}(2, 2, 2)$
Module-toggle allocation	$K_{\Omega q} = [34.8689, 27.3479, 16.8925]$, $K_q = [60.0511, 59.6435, 47.2322]$, $\alpha_f = 0.2895$	$K_{\Omega q} = [55, 40, 30]$, $K_q = [65, 65, 65]$, $\alpha_f = 0.20$
Module-toggle Backstepping gains	$c_1 = \mathrm{diag}(2.5029, 3.0133, 3.5872)$, $c_2 = \mathrm{diag}(1.8811, 1.5180, 1.5678)$; $c_3 = \mathrm{diag}(2.7003, 2.9653, 3.1537)$, $c_4 = \mathrm{diag}(1.3247, 1.4669, 1.9170)$	$c_1 = c_3 = \mathrm{diag}(4, 4, 4)$, $c_2 = c_4 = \mathrm{diag}(2, 2, 2)$
RBF basis and initial weights	Centers $[-0.8, -0.4, 0, 0.4, 0.8]$, width $b = 5.3526$, $\theta_{p,0} = 4.5024 \times 10^{-4}$, $\theta_{R,0} = 0.01185$	Centers $[-0.8, -0.4, 0, 0.4, 0.8]$, width $b = 6.0$, $\theta_{p,0} = \theta_{R,0} = 0.1$
RBF projection and output limits	$\theta_{\max} = 2.5571$, $\|\theta\|_{\max} = 5.6241$, $f_{n,\max} = 0.2575$, $\tau_{n,\max} = 0.1370$	$\theta_{\max} = 50$, $\|\theta\|_{\max} = 100$, $f_{n,\max} = 10$, $\tau_{n,\max} = 3$
RBF adaptation gains	$\gamma = [0.3101, 0.1184, 4.6998]$, $k_1 = 0.0951$, $k_{1R} = 0.0244$, $\sigma = [0.2213, 0.1743, 0.1006]$, $Q = \mathrm{diag}(58.1034, 58.1034)$, $Q_R = \mathrm{diag}(54.7321, 54.7321)$	$\gamma = [80, 20, 80]$, $k_1 = k_{1R} = 0.01$, $\sigma = [0.1, 0.05, 0.1]$, $Q = Q_R = \mathrm{diag}(50, 50)$
Error feedback gains in adaptation	$k_{p,xy} = 40.2845$, $k_{v,xy} = 16.6367$, $k_{p,z} = 41.8709$, $k_{v,z} = 31.9809$, $k_{p,R} = 28.5524$, $k_{v,R} = 18.9505$	$k_{p,xy} = 40$, $k_{v,xy} = 20$, $k_{p,z} = 40$, $k_{v,z} = 25$, $k_{p,R} = 30$, $k_{v,R} = 20$
ESO bandwidth and compensation scale	$\omega_p = 7.2199$, $\omega_R = 17.0364$, $s_p = 0.00464$, $s_R = 0.01337$	$\omega_p = \omega_R = 50$, $s_p = s_R = 1$
ESO input gains	$b_p = [0.5, 0.5, 0.5]^T$, $b_R = [0.113333, 0.173333, 0.273333]^T$	$b_p = [0.5, 0.5, 0.5]^T$, $b_R = [0.113333, 0.173333, 0.273333]^T$

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Declaration of conflicting interests

The authors declare that there is no conflict of interest.

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